



OPTOREFLECTIVE Sensor

D. MOHAN KUMAR

Here is a simple optoreflexive sensor circuit based on TCRT5000 module. It is ideal for projects like robotics, position sensing, proximity sensing, detection of reflective materials and many more. Its range is 2mm to 10mm, and the best response range is 5mm.

The circuit uses an optoreflexive sensor module that has an infrared (IR) LED and a phototransistor in a common package. The IR LED is the transmitter and the phototransistor is the receiver. The TCRT5000 in the circuit is a highly-sensitive optoreflexive sensor module that has many applications in proximity-de-

tection circuits. It gives two types of outputs (analogue and digital) based on the intensity of the reflecting IR rays falling on the phototransistor.

Analogue output is in terms of DC voltage. That means, when the phototransistor is not receiving the reflected IR rays, output is low, and when the phototransistor gets reflected IR rays, output voltage increases from low to around 3V, based on the intensity of reflected light.

The reflective sensor also has digital output. That means, when the phototransistor is not getting reflected IR rays, output remains in logic 0 (low), and when it gets reflected light, output turns to logic 1 (high). The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the optoreflexive sensor is shown in Fig. 2. Output from

the optoreflexive sensor (IC1) is used to trigger the monostable circuit built around NE555 timer (IC2.)

Resistor R1 limits the current through the internal IR LED of the sensor module. In standby mode, IR LED emits continuous IR rays. Phototransistor remains non-conducting since it is not receiving any IR rays.

Timer circuit built around IC2 also remains disabled since its trigger pin 2 is held high through resistor R4. When a reflective object comes in front of IC1 sensor module, the phototransistor inside the module conducts. This makes transistor T1 to conduct, and its collector voltage goes to ground potential, taking trigger pin 2 of IC2 also to ground.

This triggers IC2 and its output pin 3 goes high for around one second, based on the values of resistor R5 and capacitor C1. LED1 indicates activation of the timer and presence of the object in front of the optoreflexive sensor. When output of IC2 goes high, transistor T2 conducts and the piezo buzzer sounds.

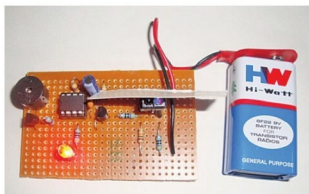


Fig. 1: Author's prototype

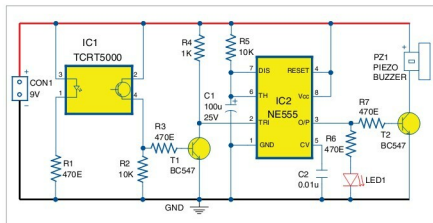


Fig. 2: Circuit diagram of the optoreflexive sensor

PARTS LIST

Semiconductors:

- IC1 - TCRT5000 optoreflexive sensor
- IC2 - NE555 timer
- T1, T2 - BC547 npn transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R6, R7 - 470-ohm
- R2, R5 - 10-kilo-ohm
- R4 - 1-kilo-ohm

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2 - 0.01 μ F ceramic disk

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 4-pin connector
- 9V PP3 battery